ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

SENSORS AND INSTRUMENTATION

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 3rd year final examination: static and dynamic measurement characteristics and errors, sensors for strain, temperature, position, speed and distance, actuators, signal conditioning, current loops and noise, data acquisition, sampling, digital filtering and discrete-time control.

Written by Academic Central for practice, following sensors, actuators and instrumentation content in South African mechatronics modules (e.g. University of Pretoria MBB 410 Mechatronics: sensors and actuators, signal conditioning, sampling and anti-aliasing, digital communication and discrete-time control). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae and substitutions. Give answers correct to TWO decimal places, with units.
3. Non-programmable scientific calculators may be used.
4. Number your answers exactly as the questions are numbered.

QUESTION 1: MEASUREMENT FUNDAMENTALS

1.1 Define the following characteristics of a sensor: accuracy, precision, resolution, sensitivity, range and hysteresis. (6)

1.2 A temperature sensor gives 0.5 V at 0 °C and 1.5 V at 100 °C, and is linear.

1.2.1 Calculate its sensitivity. (2)

1.2.2 Calculate the temperature when the output is 1.12 V. (1)

1.3 Explain the difference between systematic errors and random errors, and how each can be reduced. (2)

1.4 Five readings of a length are 10.2, 10.4, 10.1, 10.3 and 10.5 mm. Calculate the mean and the sample standard deviation. (4)

1.5 A temperature probe behaves as a first-order system with a time constant of 5 s. It is moved from air at 20 °C into an oil bath at 100 °C.

1.5.1 Calculate the probe's reading 10 s later. (3)

1.5.2 How long must the operator wait before the reading is within 2% of the final value? Explain why dynamic response matters for a sensor in a control loop. (2)

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QUESTION 2: SENSORS

2.1 A 120 Ω strain gauge with a gauge factor of 2.0 is bonded to a robot arm link and measures a strain of 500 μm/m.

2.1.1 Calculate the change in resistance. (2)

2.1.2 The gauge is one arm of a quarter bridge with a 5 V excitation. Calculate the approximate bridge output voltage. (3)

2.1.3 Why is a bridge used, and how can a second gauge compensate for temperature changes? (2)

2.2 Compare a thermocouple, an RTD and a thermistor (range, accuracy, linearity). (3)

2.3 A PT100 sensor follows R = 100(1 + 0.003 85 T) Ω. Calculate its resistance at 80 °C. (2)

2.4 A potentiometer measures a joint angle over 0° to 300° with a 5 V supply. Calculate the angle when its output is 2 V. (2)

2.5 An incremental encoder has 1 024 pulses per revolution and is read with 4× quadrature decoding. A count of 2 048 is recorded in 0.1 s. Calculate the shaft speed in r/min. (3)

2.6 Compare ultrasonic, infrared and LiDAR distance sensors for a mobile robot. (3)

2.7 Why are an accelerometer and a gyroscope combined to measure the tilt angle of a balancing robot? (3)

2.8 Explain the difference between an inductive and a capacitive proximity sensor. (2)

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QUESTION 3: ACTUATORS

3.1 Choose a DC motor, a stepper motor or a servo motor for each task and give a reason: (a) the wheels of a fast mobile robot, (b) the axes of a desktop 3D printer, (c) the steering of a model car. (3)

3.2 A 1.8° stepper motor uses 1/16 microstepping and drives a lead screw with a 2 mm pitch. Calculate the number of microsteps per revolution and the linear distance per microstep. (3)

3.3 Give ONE advantage and ONE disadvantage of pneumatic actuators compared with electric motors. (2)

3.4 A motor gives 0.2 N·m at 3 000 r/min and drives a 20:1 gearbox that is 90% efficient. Calculate the output speed and torque. (2)

3.5 An electric linear actuator lifts a 20 kg load vertically at 0.05 m/s. It is 70% efficient and runs from 24 V DC. Calculate the useful output power, the input power and the current drawn. (5)

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QUESTION 4: SIGNAL CONDITIONING

4.1 The bridge output in 2.1.2 must be amplified to 2.5 V by an instrumentation amplifier with gain G = 1 + 50 kΩ / R_G.

4.1.1 Calculate the gain needed. (2)

4.1.2 Calculate R_G. (2)

4.2 A signal contains useful frequencies up to 100 Hz and is sampled at 1 kHz. Design a first-order RC anti-aliasing filter using C = 100 nF, and state its cut-off frequency. (3)

4.3 A pressure transmitter gives a 4 to 20 mA signal for 0 to 10 bar.

4.3.1 Give TWO reasons why current loops are used in factories. (2)

4.3.2 Calculate the pressure for a current of 12 mA. (1)

4.3.3 Calculate the resistor that converts the loop current into 1 to 5 V, and the voltage at 12 mA. (2)

4.4 Describe THREE ways to reduce electrical noise picked up by sensor cables. (3)

4.5 A sensor gives −2 V to +2 V, and the ADC accepts 0 to 5 V. Find the gain and the offset of the linear scaling circuit V_out = a V_in + b. (3)

4.6 A signal of 2 V RMS has 2 mV RMS of noise on it. Calculate the signal-to-noise ratio in dB. (2)

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QUESTION 5: DATA ACQUISITION AND DIGITAL PROCESSING

5.1 A 12-bit ADC has a 0 to 3.3 V range. Calculate its resolution and its maximum quantisation error. (3)

5.2 A machine vibrates at 250 Hz and is sampled at 400 Hz. At what frequency will the vibration appear in the samples? Explain. (3)

5.3 Readings from a distance sensor are 10, 12, 30, 11 and 13 cm.

5.3.1 Calculate the outputs of a 4-sample moving-average filter for the 4th and 5th readings. (3)

5.3.2 Why would a median filter be better for this data? (2)

5.4 Write the discrete-time (difference) equation of a PID controller with sampling time T, and explain each term. (4)

5.5 Give ONE reason why the CAN bus, and not I2C, is used for sensors spread across a vehicle or large machine. (3)

5.6 A complementary filter updates a tilt angle with $\theta = 0.98(\theta + \omega_{gyro} \times dt) + 0.02\,\theta_{acc}$. Calculate the new angle when $\theta = 10°$, $\omega_{gyro} = 20$ °/s, $dt = 0.01$ s and $\theta_{acc} = 12°$. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER